



Mirjabbar Badalov / MPA-ESV 25/26L

Laboratory Report: Impact of Current Transformer Burden on Overcurrent Protection Performance

1 Circuit Diagram Description

Components and Connections:

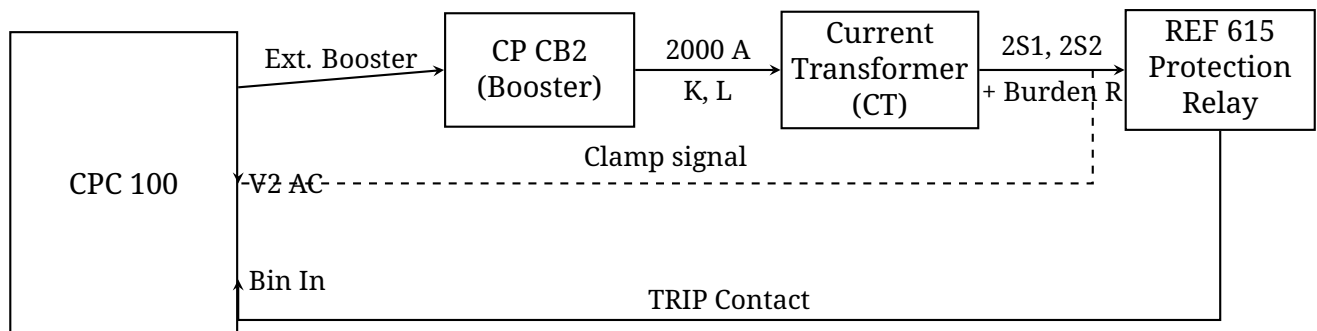


Figure 1: Simplified Circuit Diagram based on the laboratory setup.

- **CPC 100:** Primary current source and timer. Measures injected current ($V1_{AC}$) and secondary current ($V2_{AC}$).
- **CP CB2 Booster:** Amplifies injected current up to 2000 A.
- **CT (TPU 40.13):** Steps down current for the protection relay.
- **Variable Burden (R):** Simulates 10 VA (rated) and 15 VA (overload) loads.
- **REF 615 Relay:** Feeder protection relay. Trip contact feeds back to CPC 100 Bin_{In} .

2 Description of the Performed Measurements

- **Objective:** Verify effect of CT burden on saturation and tripping time.
- **Relay Setup:** REF 615, IEC Normal Inverse curve, $I_{>} = 1.0 \times I_n$ (80 A), $k = 0.1$.
- **Test Conditions:** Measured at rated burden (10 VA) and 150% burden (15 VA) across 80 A to 1210 A.

3 Processed Measured Values

The following formulas were used for the calculated columns in the tables below:

- **CT Ratio (k_I):** $80/5 = 16$
- **Ratio Error (ϵ_I):** $\epsilon_I = \frac{k_I \cdot I_{sec} - I_{prim}}{I_{prim}} \cdot 100 [\%]$
- **Theoretical Tripping Time (t_{theor}):** Calculated using the IEC Normal Inverse formula:

$$t_{theor} = \frac{k \cdot \beta}{\left(\frac{I_{prim}}{I_{>}}\right)^\alpha - 1} \quad (1)$$

Where $\alpha = 0.02$, $\beta = 0.14$, $k = 0.1$, and $I_{>} = 80 \text{ A}$.

Table 1: Measurement at Rated Burden (10 VA)

Meas.	I_{prim} (A)	I_{sec} (A)	t_{meas} (s)	φ_{sec} (°)	ϵ_I (%)	t_{theor} (s)
1	80.0	4.93	1.14	10.00	-1.43	∞
2	160.0	9.88	0.96	3.11	-1.21	1.00
3	240.0	14.83	0.86	1.94	-1.13	0.63
4	320.0	19.78	0.84	1.53	-1.08	0.49
5	400.0	24.75	0.77	1.31	-1.00	0.42
6	480.0	29.71	0.75	1.18	-0.98	0.37
7	559.8	34.63	0.85	1.08	-1.03	0.34
8	639.4	39.54	0.88	1.01	-1.06	0.32
9	718.8	44.44	0.87	0.96	-1.09	0.30
10	798.0	49.25	0.93	0.91	-1.25	0.29
11	877.0	53.85	1.29	0.88	-1.76	0.28
12	952.0	56.52	2.89	0.23	-5.00	0.27
13	1023.0	56.33	6.27	0.22	-11.90	0.26
14	1060.0	57.09	7.12	0.23	-13.83	0.25
15	1089.0	57.58	7.63	0.23	-15.40	0.25
16	1122.0	57.88	8.61	0.23	-17.47	0.25
17	1144.0	57.89	9.25	0.22	-19.03	0.24
18	1164.0	57.71	9.72	0.22	-20.67	0.24
19	1184.0	57.77	10.23	0.22	-21.94	0.24
20	1210.0	57.93	10.92	0.22	-23.40	0.24

Table 2: Measurement at Increased Burden (150% = 15 VA)

Meas.	I_{prim} (A)	I_{sec} (A)	t_{meas} (s)	φ_{sec} (°)	ϵ_I (%)	t_{theor} (s)
1	80.0	4.93	1.37	10.00	-1.46	∞
2	160.0	9.88	1.16	3.11	-1.25	1.00
3	240.0	14.83	1.04	1.94	-1.14	0.63
4	320.0	19.79	0.98	1.53	-1.06	0.49
5	400.0	24.74	0.97	1.31	-1.03	0.42
6	480.0	29.68	1.02	1.18	-1.07	0.37
7	559.8	34.61	1.00	1.09	-1.09	0.34
8	639.4	39.46	1.15	1.01	-1.26	0.32
9	719.0	44.13	1.83	0.96	-1.81	0.30
10	797.2	47.64	5.32	0.93	-4.37	0.29
11	877.2	49.60	6.66	0.91	-9.53	0.28
12	948.2	50.29	10.47	0.90	-15.13	0.27
13	965.4	48.97	13.58	0.91	-18.84	0.27
14	1013.0	49.31	15.33	0.91	-22.12	0.26
15	1051.0	49.71	16.20	0.90	-24.32	0.25
16	1060.0	49.79	17.25	0.90	-24.85	0.25
17	1094.0	49.39	18.17	0.91	-27.77	0.25
18	1111.0	49.76	18.40	0.90	-28.33	0.25
19	1149.0	49.82	19.35	0.90	-30.63	0.25
20	1181.0	49.98	20.23	0.90	-32.29	0.24

4 Graphical Results

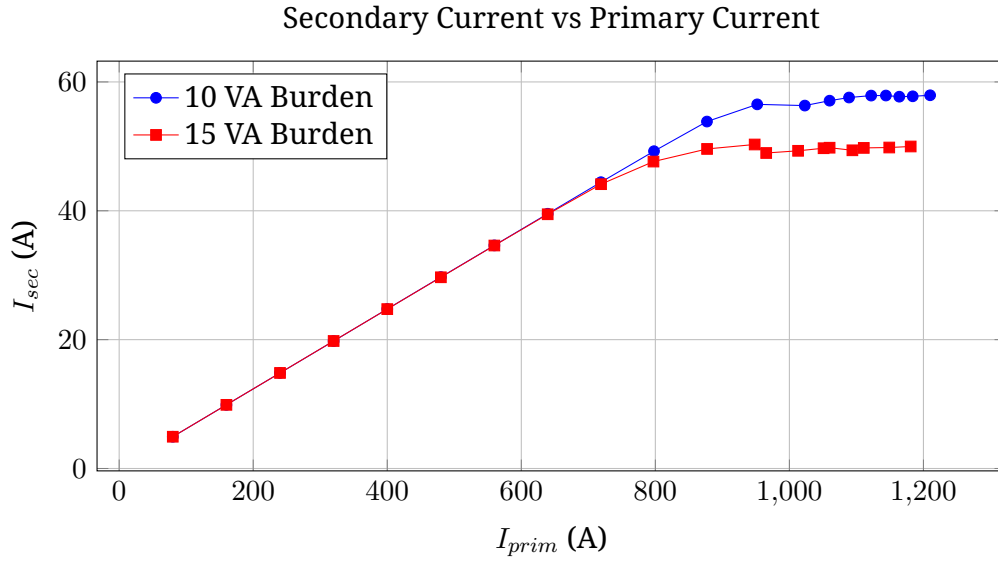


Figure 2: Secondary current output diverging due to CT saturation at different burden levels.

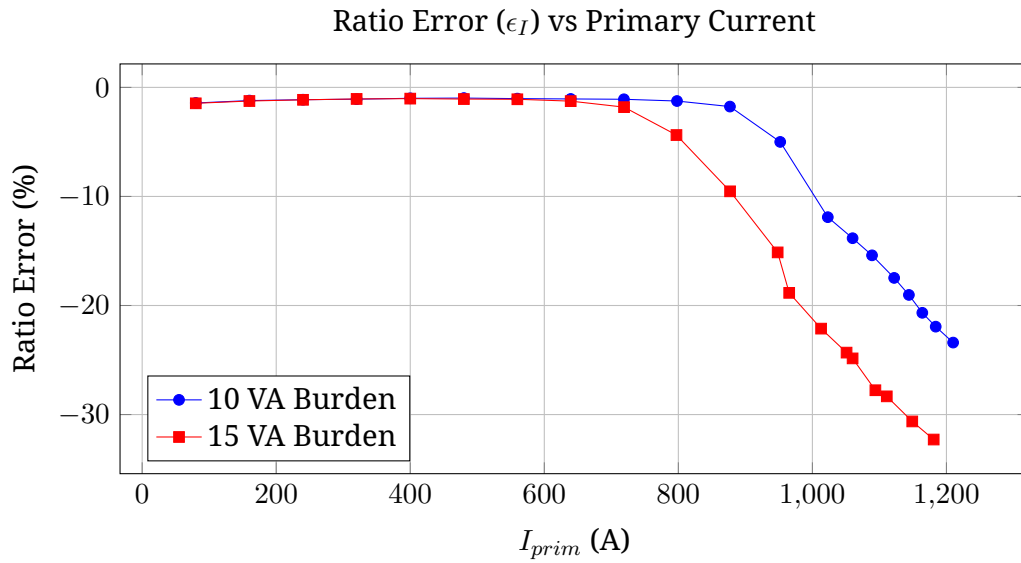


Figure 3: Ratio error demonstrating earlier and more severe saturation for the 15 VA burden.

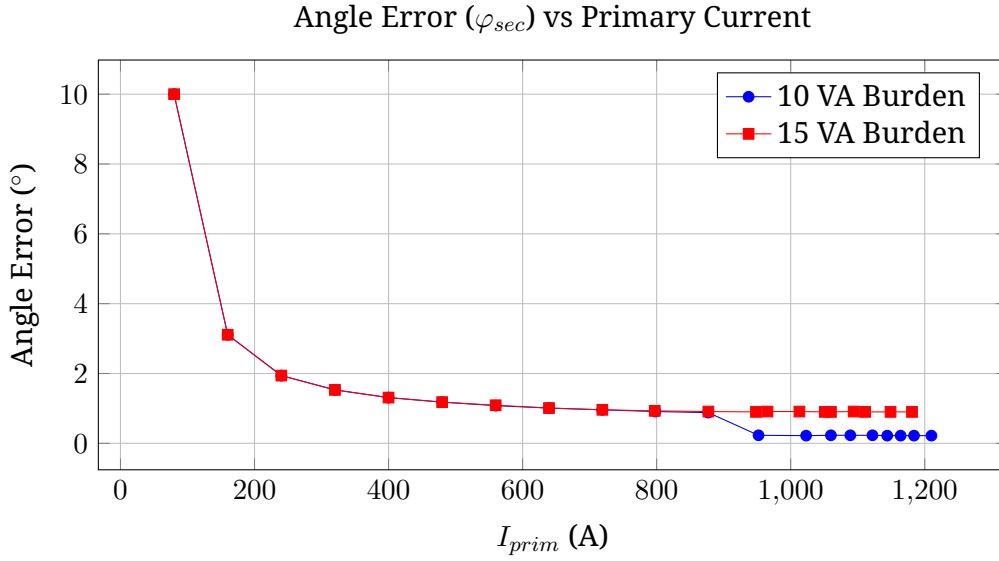


Figure 4: Angle error evolution over the tested current range.

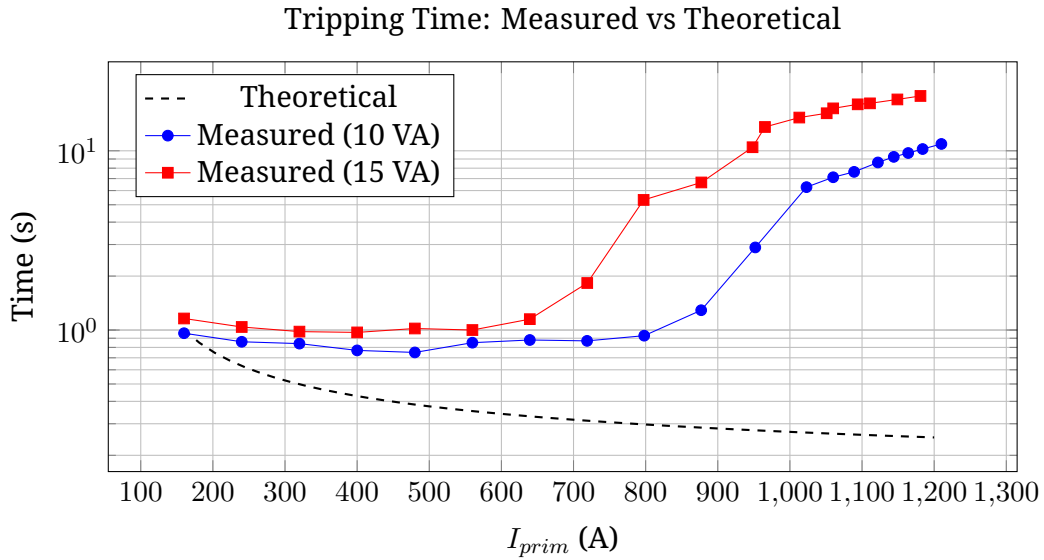


Figure 5: Impact of saturation on overcurrent protection relay tripping time. A logarithmic y-axis is used to highlight the exponential deviation at higher currents.

5 Evaluation of Results

The measurements highlight the negative impact of excessive secondary burden on CT saturation and relay performance:

- **Earlier Saturation:** At 10 VA (rated), the CT maintains $\leq 5\%$ error up to ~ 950 A. At 15 VA (overload), saturation occurs earlier, exceeding 5% error at ~ 800 A.
- **Current Limiting:** CT saturation limits the secondary current (I_{sec}), causing the relay to under-read the true fault magnitude.
- **Delayed Tripping:** Actual tripping times (t_{meas}) are severely delayed at high currents. For example, at ~ 1150 A (15 VA burden), the theoretical trip time is 0.25s, but the actual trip took over 19s.